Chapter 11 — Errors and Debugging

JavaScript reports problems with **error objects** that are **thrown** and can be **caught**. This chapter covers how errors work, how to design your own, and how to find bugs fast with the console, stack traces and debuggers — in the browser and in Node.

What you'll learn

1. The built-in error types and what every error object contains.
2. `throw`, `try` / `catch` / `finally` — exact control flow, rethrowing, traps.
3. Custom error classes and a central error-handling layer.
4. Error chaining with `cause` (ES2022) and `AggregateError`.
5. Errors in asynchronous code (preview; full story in Chapter 12).
6. Guaranteed cleanup with `finally` and `using`.
7. The console toolbox beyond `console.log`.
8. Stack traces and source maps.
9. DevTools and the Node inspector; a systematic debugging method.
10. A gallery of common runtime errors with their exact messages.

Before you start. Custom errors use classes and `extends` (Chapter 9); type coercion, the source of many silent bugs, is in Chapter 2.

All runnable code is in `code/`. Outputs are real (Node 24, V8). Error **messages** are not standardized — Firefox and Safari word them differently — but error **types** are.

11.1 Errors and the built-in error types

What it is

An **error** (or **exception**) is an object describing something that went wrong. When code **throws**, normal execution stops and the engine unwinds the **call stack** (the functions currently running, newest on top — see Chapter 12) looking for a `catch`. If none is found the error is **uncaught**: the browser logs it and stops that script task; Node prints it and exits with a non-zero **exit code** (`0` = success, anything else = failure).

- **Runtime error** — thrown while code runs (`TypeError`, `RangeError`, …).
- **Parse-time (syntax) error** — the engine can't read the code; *nothing* in that script/module runs.

The built-in types

Type	Thrown when…	Typical example
`Error`	Generic base type — use for your own errors or subclass it	`throw new Error('config missing')`
`TypeError`	A value is the wrong type for an operation	`null.length`, calling a non-function, assigning to a `const`
`ReferenceError`	A name doesn't exist, or is used in its TDZ (before its `let`/`const` line)	typo in a variable name
`SyntaxError`	Code (or JSON) can't be parsed	`JSON.parse('{bad}')`, `new Function('1 +')`
`RangeError`	A number is outside the allowed range	`new Array(-1)`, `toFixed(101)`, stack overflow
`URIError`	Malformed URI component	`decodeURIComponent('%')`
`EvalError`	Not thrown by modern engines; kept for compatibility	—
`AggregateError` (ES2021)	Several errors reported together	`Promise.any` when every promise rejects
`SuppressedError` (ES2027)	An error happened while disposing after another error	`using` declarations (§11.6)

All inherit from `Error`, so `err instanceof Error` is `true` for every built-in. Environments add their own: browsers throw `DOMException` (e.g., `AbortError`, `NotAllowedError`); Node adds `code` properties like `'ENOENT'` ("Error NO ENTry" — file not found) to system errors.

`code/01_error_types.mjs` triggers each one for real:

```
const cases = {
  TypeError:      () => null.length,
  ReferenceError: () => notDeclaredAnywhere + 1,
  SyntaxError:    () => JSON.parse('{ bad json }'),
  RangeError:     () => new Array(-1),
  URIError:       () => decodeURIComponent('%'),
  EvalError:      () => { throw new EvalError('never thrown by modern engines — only kept for compatibility'); },
  AggregateError: () => { throw new AggregateError([new Error('a'), new TypeError('b')], 'several things failed');
},
};

for (const fn of Object.values(cases)) {
  try {
    fn();
  } catch (err) {
    console.log(`${err.name.padEnd(15)} | instanceof Error: ${err instanceof Error} | ${err.message}`);
  }
}
// ...the file also prints name/message/cause/stack, Promise.any's AggregateError,
// Error.isError and a parse-time SyntaxError from new Function('return 1 +')
```

Output (first part):

```
TypeError       | instanceof Error: true | Cannot read properties of null (reading 'length')
ReferenceError  | instanceof Error: true | notDeclaredAnywhere is not defined
SyntaxError     | instanceof Error: true | Expected property name or '}' in JSON at position 2 (line 1 column 3)
RangeError      | instanceof Error: true | Invalid array length
URIError        | instanceof Error: true | URI malformed
EvalError       | instanceof Error: true | never thrown by modern engines — only kept for compatibility
AggregateError  | instanceof Error: true | several things failed
```

Anatomy of an error object

Property	Standard?	Meaning
`name`	✓ (inherited from the prototype)	The type, e.g. `'TypeError'`
`message`	✓	Human-readable description
`cause`	✓ ES2022	The underlying reason: `new Error(msg, { cause })` (§11.4)
`errors`	✓ on `AggregateError`	Array of the individual errors
`stack`	de-facto (every engine; standard proposed)	Call stack text, captured when the error is **created** (`new`), not when thrown
`code`	Node / libraries	Machine-readable code, e.g. `'ENOENT'`, `'ERR_MODULE_NOT_FOUND'`

`String(err)` / `err.toString()` gives `"Name: message"`; `Promise.any` rejecting everywhere gives `AggregateError: All promises were rejected` with `.errors`. **`Error.isError(value)`** (ES2026; Node 24 and recent browsers) is the reliable "genuine error?" check — unlike `instanceof Error` it works across **realms** (separate global environments with their own built-ins, e.g. an iframe or a Node `vm` sandbox) and can't be fooled by a hand-made prototype.

Self-check

1. `undefinedFn()` where `undefinedFn` was never declared — which error? → `ReferenceError`. If declared but `undefined`, it's a `TypeError` (not a function).
2. Which property holds the underlying error in a wrapped error? → `cause`.
3. When is a `SyntaxError` in your source file reported? → At parse time, before any of it runs.

TL;DR

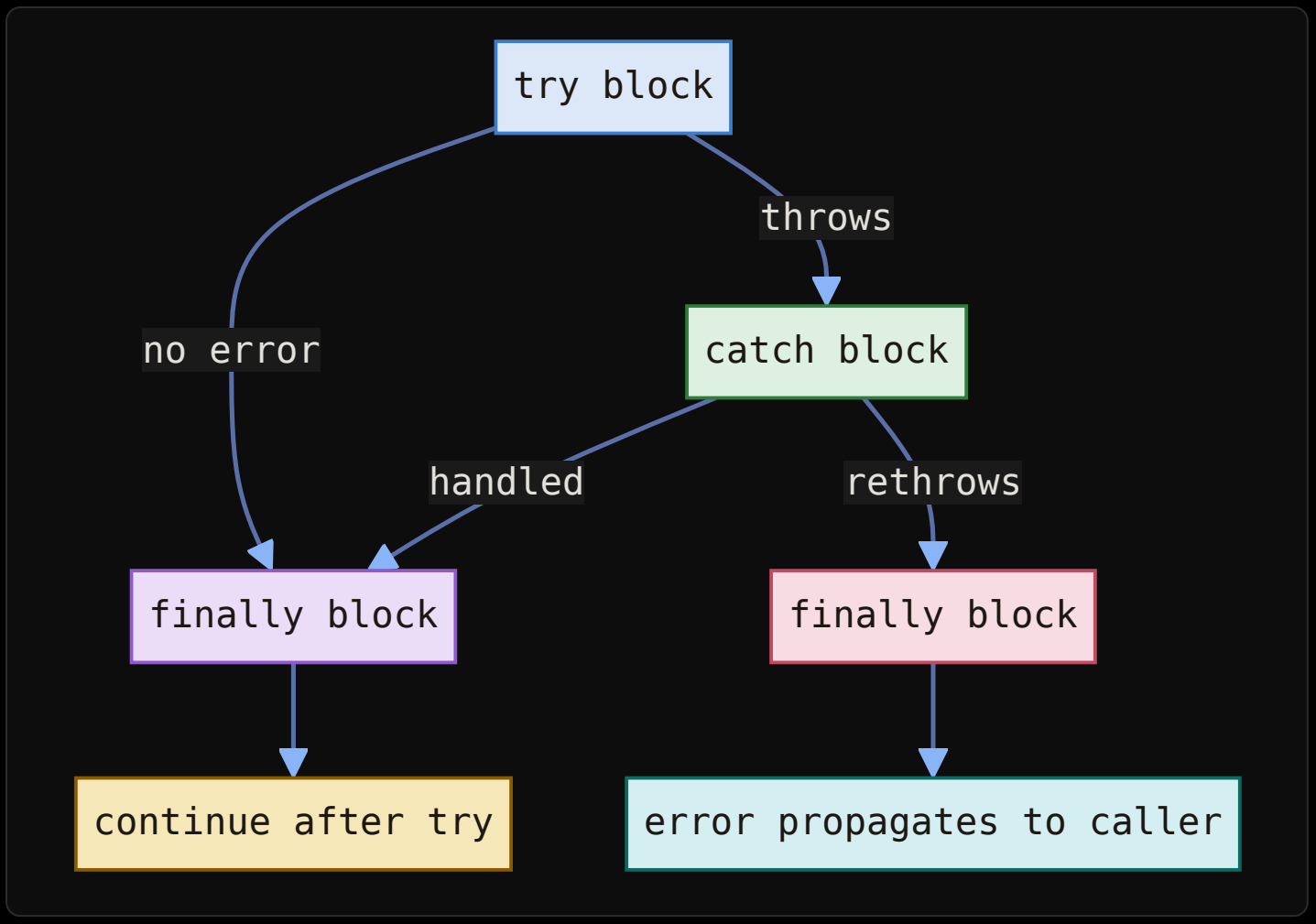
Errors are objects with `name`, `message`, `stack` (and optional `cause`). Core types: `TypeError` (wrong type), `ReferenceError` (unknown name/TDZ), `SyntaxError` (unparseable), `RangeError` (out of range), `AggregateError` (many).

11.2 `throw`, `try`, `catch`, `finally`

What it is

- `throw expression` — raises an error; execution jumps to the nearest enclosing `catch`.
- `try { … }` — code that might throw. Must be followed by `catch`, `finally`, or both.
- `catch (err) { … }` — runs only if `try` threw; `err` is the thrown value (block-scoped).
- `finally { … }` — always runs after `try`/`catch` — on success, error, or `return`. For cleanup.

How it works



If `catch` itself throws, `finally` still runs, then the new error propagates; if nothing up the stack catches it, it's uncaught.

`code/02_try_catch_finally.mjs`:

```
function parseAge(text) {
  const log = [];
  try {
    log.push('try start');
    const age = Number.parseInt(text, 10);
    if (Number.isNaN(age)) throw new TypeError(`not a number: "${text}"`);
    if (age < 0) throw new RangeError(`negative age: ${age}`);
    log.push('try end');
    return { age, log };
  } catch (err) {
    log.push(`catch ${err.name}`);
    return { age: null, log };
  } finally {
    log.push('finally');       // runs on success, on error, and even after return
  }
}
for (const input of ['42', 'abc', '-3']) {
  const { age, log } = parseAge(input);
  console.log(input.padEnd(4), '->', String(age).padEnd(5), log.join(' > '));
}

// finally's return OVERRIDES the try's return (and swallows errors) — avoid!
function sneaky() {
  try { throw new Error('lost forever'); }
  finally { return 'finally wins'; }  // eslint: no-unsafe-finally
}
console.log('\nsneaky():', sneaky());

// Optional catch binding (ES2019): omit (err) when you don't need it
function isValidJson(s) {
  try { JSON.parse(s); return true; } catch { return false; }
}
console.log('isValidJson:', isValidJson('{"a":1}'), isValidJson('{a:1}'));

// Catch only what you understand; rethrow the rest
function loadUser(raw) {
  try {
    return JSON.parse(raw).user.name;
  } catch (err) {
    if (err instanceof SyntaxError) return '(invalid JSON — using guest)';
    throw err;                    // a TypeError here means a real bug: don't hide it
  }
}
console.log('rethrow 1  :', loadUser('{oops'));
try { loadUser('{"nouser":true}'); } catch (err) { console.log('rethrow 2  :', err.name, '-', err.message); }

// (the file also throws a string, 42 and a plain object: only the Error has a .stack)

// try/catch is synchronous: it cannot catch an error thrown later by a callback
try {
  setTimeout(() => {
    try { throw new Error('from timer'); } catch (err) { console.log('\ntimer error caught INSIDE the callback:',
err.message); }
  }, 0);
} catch {
  console.log('never printed');
}
console.log('\n(outer try block already finished)');
```

Output:

```
42   -> 42    try start > try end > finally
abc  -> null  try start > catch TypeError > finally
-3   -> null  try start > catch RangeError > finally

sneaky(): finally wins
isValidJson: true false
rethrow 1  : (invalid JSON — using guest)
rethrow 2  : TypeError - Cannot read properties of undefined (reading 'name')

(outer try block already finished)

timer error caught INSIDE the callback: from timer
```

What to notice:

- `finally` ran in all three cases even though `try` and `catch` `return`: the return value is computed, `finally` runs, then the function returns.
- A `return` (or `throw`) *inside* `finally` replaces whatever `try`/`catch` was doing — `'lost forever'` vanished. ESLint's `no-unsafe-finally` flags it.
- **Catch narrowly, rethrow the rest.** `loadUser` recovers from bad JSON (`SyntaxError`); a `TypeError` means wrong data shape — a real bug that should surface.
- `catch` receives *whatever* was thrown. Strings/plain objects have no stack and fail `instanceof Error`. **Always throw `Error` objects.**
- `try/catch` only covers code that runs *synchronously inside it*. The timer callback ran later on a fresh call stack, after the outer `try` was gone.

When to use try/catch — and when not

Use it	Don't use it
At boundaries where failure is expected: parsing input, `JSON.parse`, network/file I/O, third-party code	To replace simple checks — `if (user?.name)` beats catching a `TypeError`
In a top-level handler that turns errors into a response/UI message and logs them	Wrapping every function "just in case" — hides bugs
To add context and rethrow (§11.4)	To silently swallow errors: `catch {}` with no handling

Performance: a `try` that doesn't throw costs essentially nothing; *throwing* is relatively expensive (stack capture), so don't use exceptions for normal control flow.

In the interview

> **30-second answer:** "`throw` raises an error and unwinds the stack to the nearest `catch`. `finally` always runs — after success, error, or even `return` — so it's for cleanup; but a `return` inside `finally` overrides everything and swallows errors. `try/catch` is synchronous, so it can't catch errors from callbacks that run later — for promises you `await` inside `try`, or use `.catch()`. Always throw `Error` objects so you get a stack trace."

- *Q: What does `(() => { try { return 1 } finally { return 2 } })()` return?* A: `2`.
- *Q: Does `finally` run if `catch` rethrows?* A: Yes, then the error continues outward.
- *Q: Why is `catch (e) {}` dangerous?* A: It hides every error, including bugs, making failures silent.
- *Q: `try { } finally { }` without `catch`?* A: Legal; errors still propagate, cleanup runs.
- *Q: `err` after `throw 42`?* A: The number `42` (no stack).

TL;DR

`try` → `catch` on error → `finally` always. Throw `Error` objects, catch narrowly, rethrow what you can't handle, never `return` from `finally`; try/catch is synchronous.

11.3 Custom errors

What it is

A **custom error** is an `Error` subclass for a failure in *your* domain (`ValidationError`, `NotFoundError`). Callers react to *kinds* of failure with `instanceof` (or a `code`), and it carries extra data (invalid field, HTTP status to return — see System Design Fundamentals).

How it works

1. `class MyError extends Error`; call `super(message, options)` so `message`, `stack` and `cause` are set.
2. Set `this.name` (otherwise it prints as `Error`); `new.target.name` does it for every subclass.
3. Add fields (`code`, `status`, `field`…).
4. Handle errors in **one place** (top-level handler, Express error **middleware** — a function every request passes through — or a React error boundary) that maps types to responses and treats *unknown* errors as 500s without leaking details.

`code/03_custom_errors.mjs`:

```
class AppError extends Error {
  constructor(message, options = {}) {
    super(message, { cause: options.cause });   // pass cause up to Error
    this.name = new.target.name;                  // 'ValidationError', 'NotFoundError', ...
    this.code = options.code ?? 'APP_ERROR';
    this.status = options.status ?? 500;
  }
  toJSON() { return { name: this.name, code: this.code, message: this.message }; }
}

class ValidationError extends AppError {
  constructor(field, message) {
    super(message, { code: 'VALIDATION', status: 400 });
    this.field = field;
  }
}
class NotFoundError extends AppError {
  constructor(what, id) { super(`${what} ${id} not found`, { code: 'NOT_FOUND', status: 404 }); }
}

function placeOrder(order) {
  if (!Number.isInteger(order.qty) || order.qty < 1) throw new ValidationError('qty', 'qty must be a positive
integer');
  if (order.productId !== 'p1') throw new NotFoundError('Product', order.productId);
  return { ok: true };
}

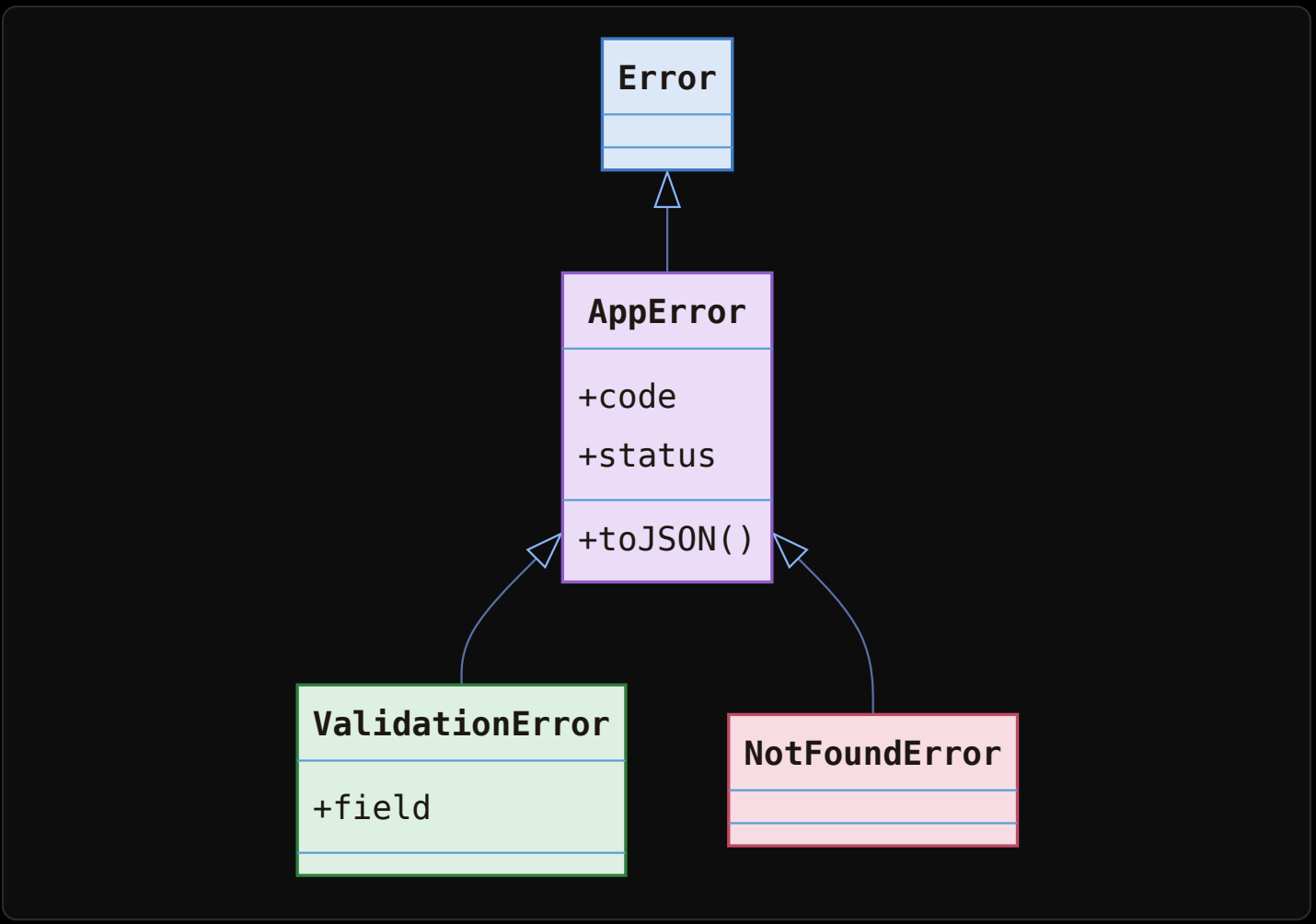
// A single place that turns errors into responses
function handle(order) {
  try {
    return { status: 200, body: placeOrder(order) };
  } catch (err) {
    if (err instanceof AppError) return { status: err.status, body: err.toJSON() };
    return { status: 500, body: { message: 'internal error' } };   // unknown -> don't leak details
  }
}
console.log(handle({ productId: 'p1', qty: 2 }));
console.log(handle({ productId: 'p1', qty: 0 }));
console.log(handle({ productId: 'p9', qty: 1 }));

const v = new ValidationError('email', 'bad email');
console.log('\ninstanceof  :', v instanceof ValidationError, v instanceof AppError, v instanceof Error);
console.log('name/field  :', v.name, v.field);
console.log('String(v)   :', String(v));
console.log('stack line 1:', v.stack.split('\n')[0]);
console.log('JSON        :', JSON.stringify(v));
console.log('plain Error JSON:', JSON.stringify(new Error('x')), '(message/stack are non-enumerable own props)');
```

Output:

```
{ status: 200, body: { ok: true } }
{
  status: 400,
  body: {
    name: 'ValidationError',
    code: 'VALIDATION',
    message: 'qty must be a positive integer'
  }
}
{
  status: 404,
  body: {
    name: 'NotFoundError',
    code: 'NOT_FOUND',
    message: 'Product p9 not found'
  }
}

instanceof  : true true true
name/field  : ValidationError email
String(v)   : ValidationError: bad email
stack line 1: ValidationError: bad email
JSON        : {"name":"ValidationError","code":"VALIDATION","message":"bad email"}
plain Error JSON: {} (message/stack are non-enumerable own props)
```



- The stack's first line says `ValidationError` because V8 builds `stack` lazily from `name` when first read. Other engines may not — always set `name` in the constructor.
- `JSON.stringify(new Error('x'))` is `{}`: `message` / `stack` are **non-enumerable** own properties (hidden from `for…in`, `Object.keys` and `JSON.stringify`) and `name` is inherited. Hence `toJSON()` (called automatically by `JSON.stringify`) — handy for logs and API bodies, but never send `stack` to end users.

Error codes vs classes

Approach	Pros	Cons
Subclasses + `instanceof`	Rich, hierarchical (`catch` a whole family)	Breaks across realms/duplicated packages (two copies of the class)
`code` string (`err.code === 'ENOENT'`)	Survives serialization, logs, process boundaries; Node's convention	No hierarchy; typos go unnoticed
Both (as above)	`instanceof` inside the app, `code` in logs and API bodies	A little more code

Common mistakes

- Forgetting `super(message)` → `ReferenceError` when you touch `this` (Chapter 9).
- Not setting `name` → logs say `Error: …` for everything.
- Leaking internals: sending `message`/`stack` of *unexpected* errors to clients can reveal SQL, file paths or secrets. Log server-side; send a generic message.
- Over-engineering: create a class only when some caller will handle it differently.

Real-world examples

Node system errors carry `code` (`ENOENT`, `EADDRINUSE` — port already taken, `ECONNREFUSED` — connection refused); `fetch` rejects with a `TypeError` on network failure and an `AbortError` `DOMException` when aborted; HTTP libraries throw errors with `status`/`response`; ORMs (libraries mapping database rows to objects) throw `UniqueConstraintError`-style subclasses.

Self-check

1. Why set `this.name`? → So `String(err)`, stacks and logs show the right type.
2. Why does `JSON.stringify(err)` print `{}`? → `message`/`stack` are non-enumerable; `name` is inherited.
3. Unknown error in a top-level handler? → Log full details internally; return a generic 500.

TL;DR

`class X extends Error`, `super(message, { cause })`, set `name`, add `code`/data, handle all errors in one place, never leak internals of unexpected errors.

11.4 Error chaining with `cause`

What it is

When you catch a low-level error and throw a higher-level one ("could not load config"), you **add context without losing the original**: since ES2022 every error constructor accepts `new Error(message, { cause })`, stored in `err.cause`. Before that, people lost the original or concatenated messages.

`code/04_error_cause.mjs`:

```
function readConfigText(name) {
  const err = new Error(`ENOENT: no such file or directory, open '${name}'`);   // pretend fs failed
  err.code = 'ENOENT';
  throw err;
}

function loadConfig(name) {
  try {
    return JSON.parse(readConfigText(name));
  } catch (err) {
    throw new Error(`Could not load config "${name}"`, { cause: err }); // add context, keep original
  }
}

function startServer() {
  try {
    loadConfig('server.json');
  } catch (err) {
    throw new Error('Server failed to start', { cause: err });
  }
}

try {
  startServer();
} catch (err) {
  let depth = 0;
  for (let e = err; e; e = e.cause) {                                    // walk the cause chain
    console.log(`${'  '.repeat(depth++)}${depth > 1 ? 'caused by: ' : ''}${e.message}${e.code ? ` [${e.code}]` :
''}`);
  }
  console.log('\nroot cause code:', rootCause(err).code);
}

function rootCause(e) { while (e.cause instanceof Error) e = e.cause; return e; }

import { inspect } from 'node:util';
const shown = inspect(new Error('outer', { cause: new Error('inner') }));
console.log('\nutil.inspect shows cause?', shown.includes('[cause]: Error: inner'));
```

Output:

```
Server failed to start
  caused by: Could not load config "server.json"
    caused by: ENOENT: no such file or directory, open 'server.json' [ENOENT]

root cause code: ENOENT

util.inspect shows cause? true
```

Each layer adds *what it was trying to do*; the root keeps *what actually failed*. Node's `console.log`/`console.error` and DevTools print the chain automatically.

`AggregateError` is the sibling for several independent failures: its `errors` array holds each one (`Promise.any` where every mirror failed; validation reporting all bad fields).

TL;DR

Wrap with `new Error('what I was doing', { cause: err })`; walk `.cause` to the root. `AggregateError` for many errors at once.

11.5 Errors in asynchronous code (preview)

The problem it solves

Async work finishes *after* the surrounding `try` has exited. Promises carry errors as **rejections** instead. The event loop and promises are explained in Chapter 12; here are the rules you need now.

`code/05_async_errors.mjs`:

```
const fetchUser = async (id) => {
  await null;                                    // simulate async work
  if (id !== 1) throw new Error(`user ${id} not found`);
  return { id, name: 'Ada' };
};

// 1) await inside try/catch — the normal way
try {
  await fetchUser(2);
} catch (err) {
  console.log('await + try/catch :', err.message);
}

// 2) .catch() on the promise
await fetchUser(3).catch((err) => console.log('.catch()          :', err.message));

// 3) Promise.allSettled — collect successes and failures together
const results = await Promise.allSettled([fetchUser(1), fetchUser(4)]);
console.log('allSettled        :', results.map((r) => (r.status === 'fulfilled' ? r.value.name :
r.reason.message)));

// 4) Forgetting to await: the try/catch finishes before the promise rejects
process.on('unhandledRejection', (reason) => {
  console.log('unhandledRejection:', reason.message, '(Node would crash here without this handler)');
});
try {
  fetchUser(5);                                  // no await!
} catch {
  console.log('never printed');
}
console.log('after the try     : the rejection has not happened yet');
await new Promise((r) => setTimeout(r, 10));      // let the event loop report it
```

Output:

```
await + try/catch : user 2 not found
.catch()          : user 3 not found
allSettled        : [ 'Ada', 'user 4 not found' ]
after the try     : the rejection has not happened yet
unhandledRejection: user 5 not found (Node would crash here without this handler)
```

Rules:

1. In an `async` function, `throw` becomes a rejected promise, and `await` of a rejected promise throws — so `try { await … } catch` works naturally.
2. **Always `await` (or `return`, or `.catch()`) every promise.** A **floating** promise's rejection escapes every `try`; `@typescript-eslint/no-floating-promises` flags them.
3. An **unhandled rejection** crashes Node by default (since Node 15) and logs "Uncaught (in promise)" in browsers.

Last-resort global handlers

Environment	Synchronous uncaught error	Unhandled promise rejection
Browser	`window.addEventListener('error', e => …)`	`window.addEventListener('unhandledrejection', e => …)`
Node	`process.on('uncaughtException', err => …)`	`process.on('unhandledRejection', reason => …)`

Use them to **log and report** (e.g., to an error tracker such as Sentry), not to carry on. After `uncaughtException` a Node process is in an unknown state — log, exit, and let a process manager (PM2, systemd, Kubernetes) restart it.

TL;DR

`await` inside `try`, or `.catch()` every promise; never leave a promise floating. Global handlers are for logging, not recovery.

11.6 Guaranteed cleanup: `finally` and `using`

What it is

Resources — files, DB connections, locks, timers, subscriptions — must be released even on failure. `finally` has always done this. **Explicit Resource Management** automates it: an object with a `[Symbol.dispose]()` method, declared with `using`, is disposed when the enclosing block exits, however it exits. It reached TC39 (the committee that designs JavaScript) stage 4 — finished — in 2025 and is slated for ES2027; supported in Node 24 and Chromium-based browsers (Chrome 134+) — check your targets for browsers. It mirrors C#'s `using`, Python's `with` and Java's try-with-resources.

`code/09_using_and_cleanup.mjs`:

```
function openFile(name) {
  console.log(`open  ${name}`);
  return {
    name,
    read() { if (name === 'bad.txt') throw new Error(`cannot read ${name}`); return `contents of ${name}`; },
    [Symbol.dispose]() { console.log(`close ${name}`); },   // called automatically by `using`
  };
}

// Classic pattern: try/finally
function classic(name) {
  const f = openFile(name);
  try { return f.read(); }
  finally { f[Symbol.dispose](); }
}
console.log('classic ->', classic('a.txt'));

// Same thing with `using`: disposed when the block exits, in reverse order, even on error
function modern(...names) {
  using first = openFile(names[0]);
  using second = openFile(names[1]);
  return [first.read(), second.read()];
}
console.log('modern  ->', modern('a.txt', 'b.txt'));
try {
  modern('c.txt', 'bad.txt');
} catch (err) {
  console.log('modern error ->', err.message, '(both files were still closed)');
}

// If disposal ALSO throws, you get a SuppressedError holding both errors
function doubleFailure() {
  using r = { [Symbol.dispose]() { throw new Error('dispose failed'); } };
  throw new Error('body failed');
}
try { doubleFailure(); } catch (err) {
  console.log(`${err.name}: error="${err.error.message}" suppressed="${err.suppressed.message}"`);
}
```

Output:

```
open  a.txt
close a.txt
classic -> contents of a.txt
open  a.txt
open  b.txt
close b.txt
close a.txt
modern  -> [ 'contents of a.txt', 'contents of b.txt' ]
open  c.txt
open  bad.txt
close bad.txt
close c.txt
modern error -> cannot read bad.txt (both files were still closed)
SuppressedError: error="dispose failed" suppressed="body failed"
```

- Disposal runs in **reverse** declaration order (last opened, first closed).
- `await using` does the same with `[Symbol.asyncDispose]()` for async cleanup.
- If body and disposal both throw you get a **`SuppressedError`**: `.error` = disposal error, `.suppressed` = the one it replaced — nothing is lost.

TL;DR

Release resources in `finally`, or with `using` + `[Symbol.dispose]` where supported. Disposal runs in reverse order, even on errors.

11.7 The console toolbox

What it is

`console` is the built-in logging object. In browsers it prints to the DevTools **Console** (filterable by level); in Node, `log/info/debug/table/dir/group/count/time` write to **stdout** and `warn/error/trace/assert` to **stderr** (a separate stream for problems, redirectable on its own).

`code/06_console_methods.mjs` runs every method below; the highlights:

```js
const users = [
  { id: 1, name: 'Ada', role: 'admin' },
  { id: 2, name: 'Linus', role: 'dev' },
];
console.table(users);

const deep = { a: { b: { c: { d: { e: 'deep' } } } } };
console.log('default depth:', deep);

for (const k of ['a', 'b', 'a', 'a']) console.count(k);
console.countReset('a');
console.count('a');

console.assert(0.1 + 0.2 === 0.3, 'assertion failed: floating point strikes again'); // prints, does NOT throw
```

Output:

```
┌─────────┬────┬──────────┬──────────┐
│ (index) │ id │ name     │ role     │
├─────────┼────┼──────────┼──────────┤
│ 0       │ 1  │ 'Ada'    │ 'admin'  │
│ 1       │ 2  │ 'Linus'  │ 'dev'    │
└─────────┴────┴──────────┴──────────┘
default depth: { a: { b: { c: [Object] } } }
a: 1
b: 1
a: 2
a: 3
a: 1
Assertion failed: assertion failed: floating point strikes again
```

Method	Use it for
`log` / `info` / `debug`	General output; separate filter levels in browsers
`warn` / `error`	Problems — yellow/red in DevTools, stderr in Node, include a stack in browsers
`%s %d %i %f %o %O %c`	Format specifiers (string, integer, float, object; `%c` = CSS styling in browsers)
`table(data, cols?)`	Arrays of objects → grid
`dir(obj, { depth })`	Full inspection (Node's `console.log` stops at depth 2 → `[Object]`)
`group` / `groupCollapsed` / `groupEnd`	Indent related messages
`count(label)` / `countReset`	"How many times did this run?"
`assert(cond, msg)`	Logs only when `cond` is false — **does not throw**
`time` / `timeLog` / `timeEnd`	Quick timing (use `performance.now()` or a profiler for real measurement)
`trace(msg)`	Print the current stack without throwing

Common mistakes

- **Browser object logs are live references**: expanding shows the *current* state, not the state at log time. Log a snapshot: `structuredClone(obj)` or `JSON.stringify(obj)`.
- Leaving debug logs in production — use `no-console` or a leveled logger (debug/info/warn/error, filter by level).
- `console.log('user: ' + user)` → `user: [object Object]`. Pass objects as separate arguments.

TL;DR

Use `table`, `dir`, `group`, `count`, `assert`, `time`, `trace` — and remember browser object logs are live.

11.8 Reading stack traces and source maps

What it is

A **stack trace** lists the function calls active when the error was created — **newest first**. Each **frame** shows the function name and `file:line:column`. Top-down tells you *where* it failed; bottom-up tells you *how you got there*.

`code/07_stack_traces.mjs` (key parts):

```
function validate(cart) { if (cart.items.length === 0) throw new Error('empty cart'); }
function checkout(cart) { validate(cart); }
function handleClick() { checkout({ items: [] }); }

try {
  handleClick();
} catch (err) {
  // shorten absolute file paths so the output is the same on every machine
  const pretty = err.stack.replace(/file:\/\/\S+\//g, '').replace(/\(.*[\\/](\S+?:\d+:\d+)\)/g, '($1)');
  console.log(pretty.split('\n').slice(0, 5).join('\n'));
}
```

Output:

```
Error: empty cart
    at validate (07_stack_traces.mjs:4:62)
    at checkout (07_stack_traces.mjs:5:27)
    at handleClick (07_stack_traces.mjs:6:26)
    at 07_stack_traces.mjs:9:3
```

How to read it:

1. **Line 1** = `name: message` — *what* happened.
2. **First frame** = where it was thrown (`validate`, line 4, column 62).
3. **Following frames** = callers. Start at the first frame in *your* code (not a library or `node:internal`).
4. **Anonymous frames** (`at 07_stack_traces.mjs:9:3`) are top-level code or unnamed functions — name your functions.

V8-only helpers (Node, Chrome, Edge): `Error.stackTraceLimit` (default 10; raise it, or run `node --stack-trace-limit=50`), and `Error.captureStackTrace(obj, fn?)`, which attaches a stack to any object (optionally hiding frames above `fn`). V8 stitches **async stack traces** across `await` points (`at async functionName`), so `async/await` traces are far more useful than callback ones.

Source maps

Production code is bundled and minified (Chapter 10), so a raw trace reads `at e (app.3f9a1c.js:1:48213)`. A **source map** (`app.js.map`) maps each generated position back to the original file/line/column. DevTools applies it when it finds `//# sourceMappingURL=`; Node with `node --enable-source-maps`. Error trackers let you upload maps privately so traces are readable without publishing source.

TL;DR

First line = *what*, first frame in your code = *where*, the rest = *how*. Source maps for bundled code; name functions for clearer traces.

11.9 Debugging with DevTools and the Node inspector

What it is

A **debugger** lets you **pause** at a chosen line, **inspect** every variable in scope, and **step** line by line — far faster than add-a-log-and-rerun for "why is this value wrong here?". Browsers ship one in **DevTools** (F12 / Cmd+Option+I); Node exposes the same Chrome DevTools protocol through its **inspector**.

Core concepts

Concept	What it does
Breakpoint	Pauses before a line runs. Click a line number (Sources panel in Chrome, Debugger in Firefox) or write `debugger;` (ignored when no debugger is attached).
Conditional breakpoint	Pauses only when an expression is true (`item.qty > 100`) — essential in loops.
Logpoint	Logs a message without pausing or editing code.
Pause on exceptions	Stops at the throwing line (optionally caught ones too) — state at the moment of failure.
Step over (F10)	Run the current line, including calls, stop at the next.
Step into (F11)	Enter the function called on this line.
Step out (Shift+F11)	Finish the current function, stop in its caller.
Resume (F8)	Continue to the next breakpoint.
Call Stack pane	The frames — click one to inspect that caller's variables.
Scope pane	Local, closure and global variables for the frame (incl. closures from Chapter 4 and `#private` fields).
Watch expressions	Re-evaluated at every pause.
Console while paused	Evaluate expressions in the paused scope, even change variables.

(Shortcuts are Chrome's; VS Code uses F10/F11/Shift+F11, F5 to continue.)

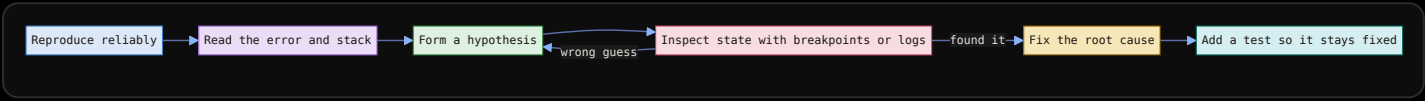
Browser-only extras: **DOM breakpoints** (pause when an element's subtree/attributes change), **event listener breakpoints** (pause on any `click`), **XHR/fetch breakpoints** (pause when a URL is requested; XHR = the older `XMLHttpRequest` API), the **Network** panel (every request, status and response — first stop for "the API call failed"), **Performance** and **Memory** panels (profiling and **heap snapshots** for finding **memory leaks** — memory never freed because something still references it), and an **ignore list** that hides library frames when stepping.

Debugging Node (not run — interactive)

```
node --inspect app.mjs          # start with the inspector on 127.0.0.1:9229
node --inspect-brk app.mjs      # same, but pause on the first line (so you can set breakpoints)
# then open chrome://inspect in Chrome -> "Open dedicated DevTools for Node"
node inspect app.mjs            # built-in terminal debugger: c(ont), n(ext), s(tep), o(ut), repl, bt
node --enable-source-maps dist/app.js   # readable traces for transpiled code
node --watch app.mjs            # restart automatically on save while you fix things
```

In **VS Code**, a "JavaScript Debug Terminal" makes editor breakpoints work for any Node script, test runner or dev server. Never expose `--inspect` on a public interface (anything but `127.0.0.1`): whoever connects can run code in your process.

A systematic method



1. **Reproduce** reliably with the smallest triggering input.
2. **Read** the actual message and stack.
3. **Hypothesize**, then **inspect** the values that confirm or refute it (breakpoints or labelled logs).

4. **Narrow down** by halving — is the value already wrong at the midpoint? (`git bisect` binary-searches commits for the change that introduced the bug.)
5. **Fix the root cause**, not the symptom.
6. **Add a test** that fails without the fix (Chapter 16).

Explaining the code line by line (even to a rubber duck) works because it forces you to state assumptions.

Worked example

`code/10_debugging_walkthrough.mjs`: a cart total that comes out wrong.

```
import assert from 'node:assert/strict';

// Data as it arrives from an HTML form or query string: everything is a string.
const cart = [
  { name: 'Book', price: '12.50', qty: '2' },
  { name: 'Pen', price: '1.25', qty: '4' },
];

// BUGGY version
function totalBuggy(items) {
  return items.reduce((sum, item) => sum + item.price * item.qty, '');
}
console.log('symptom       :', totalBuggy(cart), '(expected 30)');

// Inspect the intermediate values — labelled and structured
function totalTraced(items) {
  return items.reduce((sum, item, i) => {
    const next = sum + item.price * item.qty;
    console.log(`  step ${i}:`, { sum, typeofSum: typeof sum, line: item.price * item.qty, next });
    // debugger;   // <- with DevTools / `node inspect` attached, execution would pause here
    return next;
  }, '');
}
console.log('inspect       :');
totalTraced(cart);

// Fix: numeric initial value + explicit conversion at the boundary.
function total(items) {
  return items.reduce((sum, item) => sum + Number(item.price) * Number(item.qty), 0);
}
console.log('fixed         :', total(cart));

assert.equal(total(cart), 30);
assert.equal(total([]), 0);
assert.throws(() => assert.equal(totalBuggy(cart), 30), assert.AssertionError);
console.log('tests         : passed');
```

Output:

```
symptom       : 255 (expected 30)
inspect       :
  step 0: { sum: '', typeofSum: 'string', line: 25, next: '25' }
  step 1: { sum: '25', typeofSum: 'string', line: 5, next: '255' }
fixed         : 30
tests         : passed
```

`typeofSum: 'string'` from the first step: the initial `''` made `+` concatenate (`'25' + 5` → `'255'`), while `*` silently converted strings to numbers (Chapter 2). Nothing threw — **the most dangerous bugs are wrong values, not exceptions.** Converting at the boundary (`Number(...)`) plus tests prevents the whole class; TypeScript would also have flagged `string * string` (Chapter 16).

In the interview

> **30-second answer:** "I reproduce the bug first, read the stack trace, then use breakpoints — often conditional ones or 'pause on exceptions' — to inspect the actual state instead of guessing. I

step over/into to follow the flow and check the Network panel for API issues. For Node I use `--inspect` with Chrome DevTools or VS Code's debug terminal. Once fixed, I add a test that reproduces the bug."

- *Q: What's a logpoint?* A: A breakpoint that logs an expression instead of pausing.
- *Q: How do you debug minified production errors?* A: Source maps, uploaded to the error tracker or loaded in DevTools.
- *Q: What does `debugger` do without DevTools open?* A: Nothing.

TL;DR

Pause, inspect, step: breakpoints (conditional, logpoints, pause-on-exceptions), Scope/Watch/Call Stack, Network panel. Node: `--inspect`/`--inspect-brk` + `chrome://inspect` or VS Code. Reproduce → hypothesize → inspect → fix root cause → add a test.

11.10 Common runtime errors — a field guide

`code/08_common_runtime_errors.mjs` triggers twenty everyday mistakes (e.g. `res.data.items` on `{}`, `const [a] = {}`, `[...123]`, `new Array(2 ** 32)`, `1n + 1`) and prints V8's exact messages (Node, Chrome, Edge):

```
call a non-function         TypeError: user.getName is not a function
property of undefined       TypeError: Cannot read properties of undefined (reading 'items')
property of null            TypeError: Cannot read properties of null (reading 'textContent')
set property on undefined   TypeError: Cannot set properties of undefined (setting 'debug')
undeclared variable         ReferenceError: totl is not defined
TDZ (use before let)        ReferenceError: Cannot access 'x' before initialization
reassign const              TypeError: Assignment to constant variable.
not iterable                TypeError: {} is not iterable
spread non-iterable         TypeError: 123 is not iterable
JSON.parse(undefined)       SyntaxError: "undefined" is not valid JSON
JSON.parse(HTML)            SyntaxError: Unexpected token '<', "<!DOCTYPE html>" is not valid JSON
class without new           TypeError: Class constructor Cart cannot be invoked without 'new'
reduce empty array          TypeError: Reduce of empty array with no initial value
BigInt + Number             TypeError: Cannot mix BigInt and other types, use explicit conversions
bad array length            RangeError: Invalid array length
toFixed digits              RangeError: toFixed() digits argument must be between 0 and 100
infinite recursion          RangeError: Maximum call stack size exceeded
circular JSON               TypeError: Converting circular structure to JSON
frozen object (strict)      TypeError: Cannot assign to read only property 'a' of object '#<Object>'
in on a primitive           TypeError: Cannot use 'in' operator to search for 'x' in hello
```

What each usually means in real code

Message (V8)	Usual real-world cause	Fix
`x is not a function`	Typo in a method name; wrong import (default vs named, Chapter 10); calling a data property; library version lacks the method	Check name and import; `console.log(typeof x)`
`Cannot read properties of undefined (reading 'y')`	Data not loaded yet (async), wrong API response shape, index out of range, missing `return`	Find *which* value is `undefined` (the part before `.y`); `?.` /defaults only if absence is legitimate
`Cannot set properties of undefined`	Writing into a nested object that doesn't exist	Create it first, or `obj.cfg ??= {}`
`x is not defined`	Typo, missing import, other scope, browser-only global in Node (`window`, `document`)	Declare/import it; check the environment
`Cannot access 'x' before initialization`	`let` / `const` / `class` used in its TDZ, often via circular imports	Reorder code; break import cycles
`Assignment to constant variable.`	Reassigning a `const` or an imported binding	Use `let`, or mutate instead
`… is not iterable`	`for…of` / spread / destructuring on a plain object, `undefined`, or a number	`Object.entries(obj)`; check for `undefined`
`"undefined" is not valid JSON` / `Unexpected token '<' … is not valid JSON`	Parsing a missing value; server returned an HTML error page	Check `response.ok` and `Content-Type` before `response.json()` (Chapter 14)
`Class constructor X cannot be invoked without 'new'`	Calling a class like a function	Add `new`
`Reduce of empty array with no initial value`	`reduce` on a possibly empty array	Always pass an initial value
`Cannot mix BigInt and other types`	`bigint + number`	`BigInt(n)` or `Number(b)`
`Maximum call stack size exceeded`	Infinite/too-deep recursion; a setter assigning to itself; mutual recursion	Fix the base case; iterate instead
`Converting circular structure to JSON`	Back-references (DOM nodes, parent links)	Pick fields, or use a replacer function
`Cannot assign to read only property`	Writing to a frozen object or getter-only accessor in strict code (always on in modules/classes)	Copy instead: `{ ...obj, a: 2 }`
`Cannot use 'in' operator to search for …`	`in` on a string/number (often un-parsed JSON)	Parse first; `in` works on objects only

Also worth recognizing: `Unexpected token` / `Unexpected end of input` (missing bracket or comma), `await is only valid in async functions and the top level bodies of modules`, `Cannot use import statement outside a module` (Chapter 10), and `Failed to fetch` (a `TypeError` for network/CORS failure — Chapter 14).

Self-check

1. `Cannot read properties of undefined (reading 'map')` — what is `undefined`? → The value *before* `.map`.
2. `JSON.parse` fails with `Unexpected token '<'` after a `fetch`? → The server returned HTML (an error page or `index.html`), not JSON.
3. Why does `[].reduce((a, b) => a + b)` throw but `[].reduce((a, b) => a + b, 0)` doesn't? → With no initial value, reduce needs an element to start from.

TL;DR

Most runtime errors are `TypeError`s from `undefined`/`null` or wrong types. Read the message literally — it names the value and property. Find where the bad value came from, not just where it exploded.

Chapter 11 TL;DR

- Errors are objects (`name`, `message`, `stack`, optional `cause`); core types `TypeError`, `ReferenceError`, `SyntaxError`, `RangeError`, `URIError`, `AggregateError`, plus `SuppressedError` for `using`. `Error.isError` (ES2026) is the robust check.
- `try` → `catch` → `finally`: finally always runs; never `return` from it. Catch narrowly, rethrow the rest, throw `Error` objects. try/catch is synchronous.
- **Custom errors**: `extends Error`, `super(msg, { cause })`, set `name`, add `code`/data, handle centrally, don't leak internals.
- `cause` chains context onto the original; walk `.cause` to the root.
- **Async**: `await` in `try` or `.catch()` every promise; global `error`/`unhandledrejection` (browser) and `uncaughtException`/`unhandledRejection` (Node) are for logging.
- **Cleanup**: `finally`, or `using` + `Symbol.dispose` (ES2027, Node 24).
- **Console**: `table`, `dir`, `group`, `count`, `assert`, `time`, `trace`; browser object logs are live.
- **Stack traces**: newest frame first; source maps for bundled code.
- **Debuggers**: breakpoints (conditional, logpoints, pause on exceptions), step over/into/out, Scope/Watch/Call Stack, Network; Node `--inspect` + `chrome://inspect` or VS Code.
- **Method**: reproduce → read → hypothesize → inspect → fix root cause → add a test.

Next: the event loop, promises and `async`/`await` — Chapter 12: Asynchronous JavaScript.